



## Lab Sheet 4

# SQL Operators

## Arithmetic Operators

### 1. Perform addition of two numbers using +.

```
SELECT 10 + 20 AS Addition;
```

### 2. Perform subtraction using -.

```
SELECT 20 - 10 AS Subtraction;
```

### 3. Perform multiplication using \*.

```
SELECT 10 * 5 AS Multiplication;
```

### 4. Perform division using /.

```
SELECT 20 / 5 AS Division;
```

### 5. Calculate the total amount as Price × Quantity.

```
SELECT Price, Quantity, Price * Quantity AS TotalAmount  
FROM Products;
```

---

## Logical Operators

### 6. Demonstrate AND with two conditions.

```
SELECT *  
FROM Students  
WHERE Course = 'BCA' AND City = 'Delhi';
```

### 7. Demonstrate OR with two conditions.

```
SELECT *  
FROM Employees  
WHERE Department = 'IT' OR Department = 'HR';
```

### 8. Demonstrate NOT with a condition.

```
SELECT *  
FROM Students
```

```
WHERE NOT City = 'Delhi';
```

### **9. Find students belonging to a specific course AND city.**

```
SELECT *  
FROM Students  
WHERE Course = 'BCA' AND City = 'Delhi';
```

### **10. Find employees belonging to either of two departments.**

```
SELECT *  
FROM Employees  
WHERE Department = 'IT' OR Department = 'HR';
```

---

## **Comparison Operators**

### **11. Demonstrate the = operator.**

```
SELECT *  
FROM Students  
WHERE Course = 'BCA';
```

### **12. Demonstrate the > operator.**

```
SELECT *  
FROM Employees  
WHERE Salary > 50000;
```

### **13. Demonstrate the < operator.**

```
SELECT *  
FROM Employees  
WHERE Salary < 50000;
```

### **14. Demonstrate >= and <=.**

```
SELECT *  
FROM Employees  
WHERE Salary >= 30000 AND Salary <= 60000;
```

### **15. Demonstrate the <>/!= operator.**

```
SELECT *  
FROM Students  
WHERE City <> 'Delhi';
```

---

## **Special Operators**

### **16. Demonstrate BETWEEN for salary range.**

```
SELECT *
FROM Employees
WHERE Salary BETWEEN 30000 AND 60000;
```

### **17. Demonstrate IN for selecting multiple cities.**

```
SELECT *
FROM Students
WHERE City IN ('Delhi', 'Mumbai', 'Pune');
```

### **18. Demonstrate NOT IN.**

```
SELECT *
FROM Students
WHERE City NOT IN ('Delhi', 'Mumbai');
```

### **19. Demonstrate LIKE for names beginning with a particular letter.**

```
SELECT *
FROM Students
WHERE Name LIKE 'A%';
```

### **20. Demonstrate IS NULL and IS NOT NULL.**

#### **IS NULL:**

```
SELECT *
FROM Employees
WHERE ManagerID IS NULL;
```

#### **IS NOT NULL:**

```
SELECT *
FROM Employees
WHERE ManagerID IS NOT NULL;
```

---

## **Set Operations**

### **21. Demonstrate UNION using two compatible queries.**

```
SELECT City
FROM Students
UNION
SELECT City
FROM Employees;
```

### **22. Demonstrate UNION ALL.**

```
SELECT City
FROM Students
UNION ALL
SELECT City
FROM Employees;
```

## 23. Demonstrate INTERSECT where supported by the DBMS.

```
SELECT City
FROM Students
INTERSECT
SELECT City
FROM Employees;
```

## 24. Demonstrate EXCEPT/MINUS where supported.

### EXCEPT:

```
SELECT City
FROM Students
EXCEPT
SELECT City
FROM Employees;
```

### For MySQL:

```
SELECT DISTINCT City
FROM Students
WHERE City NOT IN (
    SELECT City
    FROM Employees
);
```

---

## Problem using Multiple Operators

**Perform a suitable problem using multiple operators together, such as Arithmetic + Logical + Comparison + Special operation.**

```
SELECT Name, Department, Salary,
       Salary * 1.10 AS SalaryWithBonus
FROM Employees
WHERE (Department = 'IT' OR Department = 'HR')
AND Salary BETWEEN 30000 AND 80000
AND Name LIKE 'A%';
```

## OUTPUT:-

### Arithmetic Operators

#### 1. Perform addition of two numbers using +.

```
Addition
-----
30
```

#### 2. Perform subtraction using -.

Subtraction  
-----  
10

### 3. Perform multiplication using \*.

Multiplication  
-----  
50

### 4. Perform division using /.

Division  
-----  
4

### 5. Calculate the total amount as Price × Quantity.

| Price | Quantity | TotalAmount |
|-------|----------|-------------|
| ----- | -----    | -----       |
| 35.50 | 2        | 71.00       |
| 40.50 | 1        | 40.50       |
| 50.00 | 3        | 150.00      |

---

## Logical Operators

### 6. Demonstrate AND with two conditions.

| StudentID | Name | Course | City  |
|-----------|------|--------|-------|
| -----     | ---- | -----  | ----  |
| 1         | Amit | BCA    | Delhi |

### 7. Demonstrate OR with two conditions.

| EmployeeID | Name  | Department |
|------------|-------|------------|
| -----      | ----  | -----      |
| 1          | Rahul | IT         |
| 2          | Priya | HR         |

### 8. Demonstrate NOT with a condition.

| StudentID | Name  | Course | City   |
|-----------|-------|--------|--------|
| -----     | ----  | -----  | ----   |
| 2         | Priya | BCA    | Mumbai |
| 3         | Rahul | BBA    | Pune   |

### 9. Find students belonging to a specific course AND city.

| StudentID | Name | Course | City  |
|-----------|------|--------|-------|
| -----     | ---- | -----  | ----  |
| 1         | Amit | BCA    | Delhi |

## 10. Find employees belonging to either of two departments.

| EmployeeID | Name  | Department |
|------------|-------|------------|
| -----      | ----  | -----      |
| 1          | Rahul | IT         |
| 2          | Priya | HR         |

---

## Comparison Operators

### 11. Demonstrate the = operator.

| StudentID | Name  | Course |
|-----------|-------|--------|
| -----     | ----  | -----  |
| 1         | Amit  | BCA    |
| 2         | Priya | BCA    |

### 12. Demonstrate the > operator.

| EmployeeID | Name  | Salary |
|------------|-------|--------|
| -----      | ----  | -----  |
| 1          | Rahul | 60000  |
| 3          | Amit  | 75000  |

### 13. Demonstrate the < operator.

| EmployeeID | Name  | Salary |
|------------|-------|--------|
| -----      | ----  | -----  |
| 2          | Priya | 40000  |
| 4          | Neha  | 35000  |

### 14. Demonstrate >= and <=.

| EmployeeID | Name  | Salary |
|------------|-------|--------|
| -----      | ----  | -----  |
| 1          | Rahul | 50000  |
| 2          | Priya | 40000  |
| 3          | Amit  | 60000  |

### 15. Demonstrate the <>/!= operator.

| StudentID | Name  | Course | City   |
|-----------|-------|--------|--------|
| -----     | ----  | -----  | ----   |
| 2         | Priya | BCA    | Mumbai |
| 3         | Rahul | BBA    | Pune   |

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## Special Operators

### 16. Demonstrate BETWEEN for salary range.

| EmployeeID | Name | Salary |
|------------|------|--------|
| -----      | ---- | -----  |

|   |       |       |
|---|-------|-------|
| 1 | Rahul | 50000 |
| 2 | Priya | 40000 |
| 3 | Amit  | 60000 |

### 17. Demonstrate IN for selecting multiple cities.

| StudentID | Name  | City   |
|-----------|-------|--------|
| 1         | Amit  | Delhi  |
| 2         | Priya | Mumbai |
| 3         | Rahul | Pune   |

### 18. Demonstrate NOT IN.

| StudentID | Name  | City   |
|-----------|-------|--------|
| 3         | Rahul | Pune   |
| 4         | Neha  | Jaipur |

### 19. Demonstrate LIKE for names beginning with a particular letter.

| StudentID | Name   |
|-----------|--------|
| 1         | Amit   |
| 5         | Anjali |

### 20. Demonstrate IS NULL and IS NOT NULL.

#### IS NULL:

| EmployeeID | Name | ManagerID |
|------------|------|-----------|
| 4          | Neha | NULL      |

#### IS NOT NULL:

| EmployeeID | Name  | ManagerID |
|------------|-------|-----------|
| 1          | Rahul | 101       |
| 2          | Priya | 102       |
| 3          | Amit  | 101       |

## Set Operations

### 21. Demonstrate UNION using two compatible queries.

| City   |
|--------|
| Delhi  |
| Mumbai |
| Pune   |
| Jaipur |

## 22. Demonstrate UNION ALL.

```
City
----
Delhi
Mumbai
Pune
Delhi
Mumbai
```

## 23. Demonstrate INTERSECT where supported by the DBMS.

```
City
----
Delhi
Mumbai
```

## 24. Demonstrate EXCEPT/MINUS where supported.

```
City
----
Pune
Jaipur
```

---

## Problem using Multiple Operators

**Perform a suitable problem using multiple operators together, such as Arithmetic + Logical + Comparison + Special operation.**

| Name   | Department | Salary | SalaryWithBonus |
|--------|------------|--------|-----------------|
| ----   | -----      | -----  | -----           |
| Amit   | IT         | 50000  | 55000           |
| Anjali | HR         | 60000  | 66000           |